



Unit 4 · Outcome 1 · Climate Change

Reading the sky's diary: measuring atmospheric change

KK07

VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.



The atmosphere doesn't keep receipts — so how do we know?

The big idea The atmosphere doesn't keep receipts — so how do we know? Direct, reliable thermometer and gas readings only go back about 170 years . The atmosphere itself is constantly mixing and doesn't store a tidy record of what it used to be. Yet scientists confidently talk about CO₂ levels 800,000 years ago . How? This dot point is really a detective story . The crime scene is the climate of the deep past, and there's no eyewitness — no one stood in Antarctica in 130,000 BCE with a CO₂ analyser. So scientists do what detectives do: they read the evid

The instrumental record — sharp, but short

Measuring the present The instrumental record — sharp, but short Start with what we can measure right now . Two iconic stations define the modern atmospheric record, and one of them is on Australian soil. Kennaook / Cape Grim, NW Tasmania Since April 1976 , the Kennaook / Cape Grim Baseline Air Pollution Station has continuously sampled the cleanest air on Earth — air that has blown thousands of kilometres across the Southern Ocean with no recent human influence. It's one of only three World Meteorological Organization "super-stations" and is the Souther

The sky's diary, frozen one year at a time

Ice-core sampling The sky's diary, frozen one year at a time This is the showpiece of palaeoclimate science. Each winter, snow falls on the Antarctic ice sheet and never melts. New snow buries the old, squeezing it into firn and then solid ice — and as it seals, it traps tiny bubbles of the actual atmosphere from that year. Drill down, and you drill back in time : deeper ice is older ice. Scene 1 · The ice-core time machine Drag the depth slider to descend through an Antarctic core. Watch age, CO₂ and temperature change with depth. Three.js · 3D Depth 0

What's measured	Direct or proxy?	What it tells us
Trapped air bubbles (gas extracted in the lab)	Direct	Actual past CO ₂ , CH ₄ and N ₂ O concentrations
Water isotopes (δD, δ ¹⁸ O in the ice)	Proxy	Past air temperature when the snow fell
Annual layers (visible / chemical banding)	Dating	The age of each slice of ice
Dust & ash bands	Proxy	Aridity, wind strength, volcanic eruptions

The other recorders: rings, reefs and shells

Palaeoclimate records The other recorders: rings, reefs and shells Ice cores need ice. For places and times without it, scientists read other natural archives, each with its own reach and resolution. The skill VCAA tests is matching the right proxy to the timescale — and knowing every one of these is indirect. Scene 2 · The proxy archive explorer Click each archive to see how far back it reaches and how fine its detail is — the reach-vs-resolution trade-off. Three.js · 3D Archive Tap one → Reaches back — Resolution — Type — Click an archive object to i

Archive	What records the signal	Reaches back	Climate inferred
Tree rings (dendroclimatology)	Annual ring width & density	~10,000 yr	Warmth & rainfall of each growing season
Coral cores	Density bands & isotopes in the skeleton	~100s–1,000s yr	Sea-surface temperature, ocean chemistry
Ice cores	Trapped air + water isotopes	~800,000 yr	Gas levels (direct) + temperature (proxy)
Ocean sediment + foraminifera	$\delta^{18}\text{O}$ in microscopic shells	millions of yr	Deep-time temperature & ice volume
Pollen & speleothems	Plant types in mud; cave-mineral layers	1,000s–100,000s yr	Past vegetation, rainfall, temperature

The present-day watch network

Atmosphere & ocean monitoring The present-day watch network Bring it back to now. To track present atmospheric and ocean change, the planet is wrapped in a monitoring network. Spin the globe and toggle between the two layers the dot point names. Scene 3 · Earth's monitoring network Drag to spin. Toggle the atmospheric stations and the ocean float array. Three.js · 3D Layer Atmosphere Watching CO_2 & air temp Station Cape Grim Drag to spin · toggle layers below Atmosphere Ocean Pause spin Orange pins = baseline air stations (Cape Grim, Mauna Loa) Switch t

Command words for this dot point

◆ COMMAND WORDS

Decode the question Command words for this dot point The verb tells you what to do. Tap each card — and note the typical mark range, because it signals how much to write. Describe 2–3 reveal Distinguish 2–4 reveal Explain 3–4 reveal Analyse 4–6 reveal Evaluate 4–6 reveal Justify 2–3 reveal

Six traps that quietly cost marks

⚠ EXAM TRAPS

Don't lose easy marks Six traps that quietly cost marks Trap 1 · "Ice cores measure temperature directly" No. The trapped air measures gas composition directly; temperature is read from a water-isotope proxy. Mixing these up is the most common error on this topic. Trap 2 · Calling every proxy "direct" Tree rings, coral and foraminifera are all indirect/proxy — you infer climate from a biological or chemical response. Only the ice-core gas is a direct sample of the past atmosphere. Trap 3 · Deeper ice = younger The opposite. Snow buries snow, so deeper

The P·E·L trainer

P·E·L WRITING

Write like a state-ranker The P·E·L trainer For "explain / analyse / evaluate" answers, structure earns marks. P·E·L = Point → Evidence → Link . Tap each step for the move and a worked example for this dot point. P·E·L · build a band-6 paragraph on measuring past climate P Point E Evidence L Link



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

Same question, two answers

See the difference Same question, two answers Q. Explain how ice-core sampling allows scientists to measure past atmospheric CO₂ concentrations. (3 marks) Weak response "Scientists drill into the ice in Antarctica and study it. The ice is very old so it can tell them what the climate used to be like and how much CO₂ there was a long time ago." No mechanism. How does ice "tell them" the CO₂? Never mentions trapped air. This is a description of what they do , not an explanation of how it works . Marks: 1 / 3 — only "old ice = past climate" credited Strong



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

Practice questions · 22 marks

★ PRACTICE & SELF-MARK

Now you Practice questions · 22 marks Try each in your head or on paper first. Then reveal the weak answer, the strong answer, and the marking guide — and self-mark honestly . Your running total tracks bottom-right. Q1 Distinguish between a direct measurement and a proxy measurement of past climate. Command: Distinguish · use a linking word 2 marks Reveal answers ▼ ✕ Weak "A direct measurement is taken directly and a proxy is taken using a proxy." Circular — restates the words without explaining them. 0 marks. ✓ Strong "A direct measurement uses an instr

One-screen recap

♦ RECAP / CHEAT SHEET

Lock it in One-screen recap If you remember five things Direct vs proxy is the master distinction. Direct = instrument reads the variable itself; proxy = infer climate from a natural recorder. Ice cores are both: trapped air = direct past gas levels; water isotopes = proxy for past temperature. Deeper = older. Match the archive to the timescale: tree rings/coral (high-res, short reach) → ice cores (~800,000 yr) → ocean sediment + foraminifera (millions of yr). Present-day monitoring is atmospheric AND ocean — Cape Grim & Mauna Loa for gases; Argo floats

★ Good vs bad worked answers

Study the contrast — the same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer structure — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

Scientists drill into the ice in Antarctica and study it. The ice is very old so it can tell them what the climate used to be like and how much CO₂ there was a long time ago."

No mechanism. How does ice "tell them" the CO₂?

Never mentions trapped air. This is a description of what they do, not an explanation of how it works. Marks: 1 / 3 — only "old ice = past climate" credited

✓ STRONG / MODEL ANSWER

As snow falls and is buried each year, it compresses into ice and traps bubbles of the atmosphere from that time. Because deeper ice is older, drilling a core recovers a dated sequence. Scientists then extract and analyse the trapped air, giving a direct measurement of past CO₂ concentration." Full mechanism chain, correct direction (deeper = older), and the key word "direct". Each mark earned cleanly. Marks: 3 / 3

✗ WEAK ANSWER

A direct measurement is taken directly and a proxy is taken using a proxy." Circular — restates the words without explaining them. 0 marks.

✓ STRONG / MODEL ANSWER

A direct measurement uses an instrument to record the variable itself, such as a gas analyser reading present CO₂, whereas a proxy measures a natural feature that responded to climate — such as tree-ring width — and the climate is inferred indirectly from it." Both halves defined, linked with "whereas", with an example each.

✗ WEAK ANSWER

The ice has air and chemicals in it that show the temperature and the gases from the past." Gestures at both but explains neither, and blurs the direct/proxy distinction. ~1 mark.

✓ STRONG / MODEL ANSWER

As snow is buried it traps bubbles of ancient air; extracting and analysing this gas gives a direct measurement of past CO₂, CH₄ and N₂O. Separately, the ratio of water isotopes ($\delta^{18}\text{O}$ / δD) in the ice depends on the temperature when the snow fell, acting as a proxy for past air temperature. Because both are recorded in the same dated core, the two records can be compared layer by layer." Cleanly separates the direct gas record from the proxy temperature record.

✗ WEAK ANSWER

No, tree rings are not the best because ice cores are better and go back further." Right verdict, no reasoning. Doesn't use reach/resolution. ~1 mark.

✓ STRONG / MODEL ANSWER

The claim is incorrect. Tree rings have high resolution (annual) but only reach ~10,000 years, so they cannot record climate 500,000 years ago. For that timescale an ice core (up to ~800,000 yr) or ocean-sediment record (millions of yr) is required. Tree rings would be the better choice for fine detail over recent millennia, but not for deep time — the archive must match the timescale." Verdict + reach figures + resolution + correct alternative + the matching principle.

✗ WEAK ANSWER

Ice cores are good because they go back a long way and tell us about CO₂. They are bad because they are hard to drill. Overall they are useful." Touches strength and limitation but both are thin and the "limitation" (hard to drill) isn't a scientific limitation of the data. No use of evidence. ~2 marks.

✓ STRONG / MODEL ANSWER

Ice cores are a powerful method because the trapped air gives a direct measurement of past greenhouse gases stretching back ~800,000 years (EPICA Dome C), and water isotopes add a temperature proxy — together revealing the tight CO₂–temperature coupling across eight ice-age cycles. A limitation is that resolution decreases with depth/age as deep layers compress, so very old sections are less finely dated than recent ones; ice cores are also restricted to polar regions with permanent ice. On balance, for long-term, global atmospheric composition they are the single most valuable method, best combined with other proxies to cross-check the record." Two genuine scientific strengths, a real limit